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- Let u and v be some primitive type variables and α and β be some values that match the types of u and v (u and v might be the same variable, α and β might be the same value). When does $\sigma[u \mapsto \alpha][v \mapsto \beta] = \sigma[v \mapsto \beta][u \mapsto \alpha]$ and when does $\sigma[u \mapsto \alpha][v \mapsto \beta] \equiv \sigma[v \mapsto \beta][u \mapsto \alpha]$? Discuss the four different cases depending on whether $u \equiv v$ and whether $\alpha = \beta$.

		$\sigma[u \mapsto \alpha][v \mapsto \beta] = \sigma[v \mapsto \beta][u \mapsto \alpha]$?	$\sigma[u \mapsto \alpha][v \mapsto \beta] \equiv \sigma[v \mapsto \beta][u \mapsto \alpha]$?
$u \equiv v$	$\alpha = \beta$		
$u \equiv v$	$\alpha \neq \beta$		
$u \neq v$	$\alpha = \beta$		
$u \neq v$	$\alpha \neq \beta$		

- Consider state $\sigma = \{x = 2, y = 4\}$. Answer the following questions and show your work.
 - Find state $\sigma[x \mapsto \sigma(y)][y \mapsto \sigma(x)]$.
 - Find state $\sigma[x \mapsto \sigma(y)][x \mapsto \sigma[x \mapsto \sigma(y)](x - y)]$.
 - Let $\tau = \sigma[x \mapsto 3]$, and $\gamma = \tau[y \mapsto \tau(x) * 4]$. Find γ .
- Decide true or false for each of the following satisfaction. Justify your answers briefly.
 - $\{x = 4, y = 2\} \models x < 3 \rightarrow y > 2$
 - $\{x = 3, b = (2, 5, 4, 8)\} \models x > 1 \wedge \exists 0 \leq x < 4. b[x] < 3$
 - $\sigma \models \exists y. \forall x. 0 \leq x \leq 1 \rightarrow b[x] = y$, where $\sigma(b) = (2, 2, 3, 3)$
- Translate the following Java programs into statements in our programming language.
 - int** $x = 0$; **for** (**int** $i = 0$; $i < b.length$; $i++$) $\{x += b[i];\}$
 - while** ($x \neq 1$) $\{\text{if } (x \% 2 == 0) \{x = x/2;\} \text{ else } \{x++;\}\}$
 - int** $m = 8, p = 1, y = 1$; **while** ($++m < 20$) $\{p = p * (y++);\}$
- Evaluate each of the following configurations to completion. Do not use $\rightarrow^*$ or $\rightarrow^n$ in your solutions.
 - $\langle \text{if } x < 2 \text{ then } x := y + 1; w := x + 2 \text{ fi}, \sigma \rangle$, where $\sigma(x) = \sigma(y) = 3$ and $\sigma(w) = 4$.
 - $\langle x := y + 1; y := x + 1, \sigma \rangle$. Here, variables x and y are both defined in state σ .
 - $\langle W, \sigma \rangle$, where $W \equiv \text{while } x > y \text{ do } S \text{ od}, S \equiv \text{if } x > 0 \text{ then } x := x + 1 \text{ else } y := -2 * x \text{ fi}$, and $\sigma \models y < x \leq 0$.
- Let $W \equiv \text{while } x < 2 \text{ do } S \text{ od}$, where $S \equiv x := x + 1; p := p \wedge (x > 1)$.
 - Calculate $M(S, \sigma_1)$. Here, σ_1 is a state with variables x and p defined.
 - Calculate $M(W, \sigma_2)$, where $\sigma_2(x) = 4$ and $\sigma_2(p) = T$.
 - Calculate $M(W, \sigma_3)$, where $\sigma_3 \models (x = 1) \wedge p$.

7. Let $W \equiv \text{while } x > 0 \text{ do } S \text{ od}$, where $S \equiv \text{if } x < y \text{ then } x := y/x \text{ else } x := x - 1; y := b[y] \text{ fi}$. Answer the following questions.
- Calculate $M(S, \sigma_1)$ where $\sigma_1(x) = -2$ and $\sigma_1(y) = -1$.
 - Calculate $M(W, \sigma_2)$ where $\sigma_2 = \{x = 2, y = 2, b = (0, 1, 2)\}$.
 - Calculate $M(W, \sigma_3)$ where $\sigma_3 = \{x = 8, y = 2, b = (4, 2, 0)\}$.

8. In Lecture 5, without the consideration of $\perp$, we have the following conclusions:

$$\sigma \not\models \exists x \in S. p \Leftrightarrow \sigma \models \forall x \in S. \neg p$$

$$\sigma \not\models \forall x \in S. p \Leftrightarrow \sigma \models \exists x \in S. \neg p$$

We used “ $\sigma \not\models p \Leftrightarrow \sigma \models \neg p$ ” during the proof to get the above conclusions, so they are not totally correct anymore with the consideration of $\perp$. Let’s create some conclusions without replacing $\sigma \not\models p$ with $\sigma \models \neg p$. Remember that the following definitions about satisfaction of quantified predicates are still correct (I rephrased them here):

- $\sigma \models \exists x \in S. p$ means there exists $\alpha \in S$, it is the case that $\sigma[x \mapsto \alpha] \models p$.
- $\sigma \models \forall x \in S. p$ means for every value $\alpha \in S$, it is the case that $\sigma[x \mapsto \alpha] \models p$.

Fill in the blanks as required so that each of the following sentences is correct.

- Fill type 1 with one of the words: “some”, “every” or “this”.
 - Fill type 2 with either “ $\models$ ” or “ $\not\models$ ”.
- $\sigma \models \exists x \in S. p$ means for type 1 state σ and for type 1 $\alpha \in S$, it is the case that $\sigma[x \mapsto \alpha]$ type 2 p .
 - $\sigma \models \forall x \in S. p$ means for type 1 state σ and for type 1 $\alpha \in S$, it is the case that $\sigma[x \mapsto \alpha]$ type 2 p .
 - $\sigma \not\models \exists x \in S. p$ means for type 1 state σ and for type 1 $\alpha \in S$, it is the case that $\sigma[x \mapsto \alpha]$ type 2 p .
 - $\sigma \not\models \forall x \in S. p$ means for type 1 state σ and for type 1 $\alpha \in S$, it is the case that $\sigma[x \mapsto \alpha]$ type 2 p .
 - $\models \exists x \in S. p$ means for type 1 state σ , it is the case that σ type 2 $\exists x \in S. p$.
 - $\models \forall x \in S. p$ means for type 1 state σ , it is the case that σ type 2 $\forall x \in S. p$.
 - $\not\models \exists x \in S. p$ means for type 1 state σ , it is the case that σ type 2 $\exists x \in S. p$.
 - $\not\models \forall x \in S. p$ means for type 1 state σ , it is the case that σ type 2 $\forall x \in S. p$.